Open Standard To AI Factories

DCBBS, Modular Design, and the latest in Liquid Cooling

Magelli Roxas | Sr Rack PM

Steven Huang | Product Manager

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DCBBS → MODULAR DESIGN→ LIQUID COOLING

The Building Block Architecture

Modular by design — deploy at any scale, expand seamlessly

SYSTEM LEVEL

Individual building blocks customized to your workloads

RACK LEVEL

Systems integrated, validated and tested before shipping

DATA CENTER LEVEL

Multi-rack clusters deployed as a complete solution



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Supermicro End-to-End Solutions for AI Data Centers

Inside the Data Center



Supermicro Liquid-Cooled SuperCluster & Liquid Cooling Solution (Secondary)



Compute, Network & Storage Rack



Cabling Service



L2A Sidecar



In-Rack CDU



In-Row CDU

CDU Series



Secondary Liquid Loop Piping Service

Outside the Data Center

Supermicro Liquid Cooling Solution (Primary)







Cooling Tower



Primary Liquid Loop Piping Service



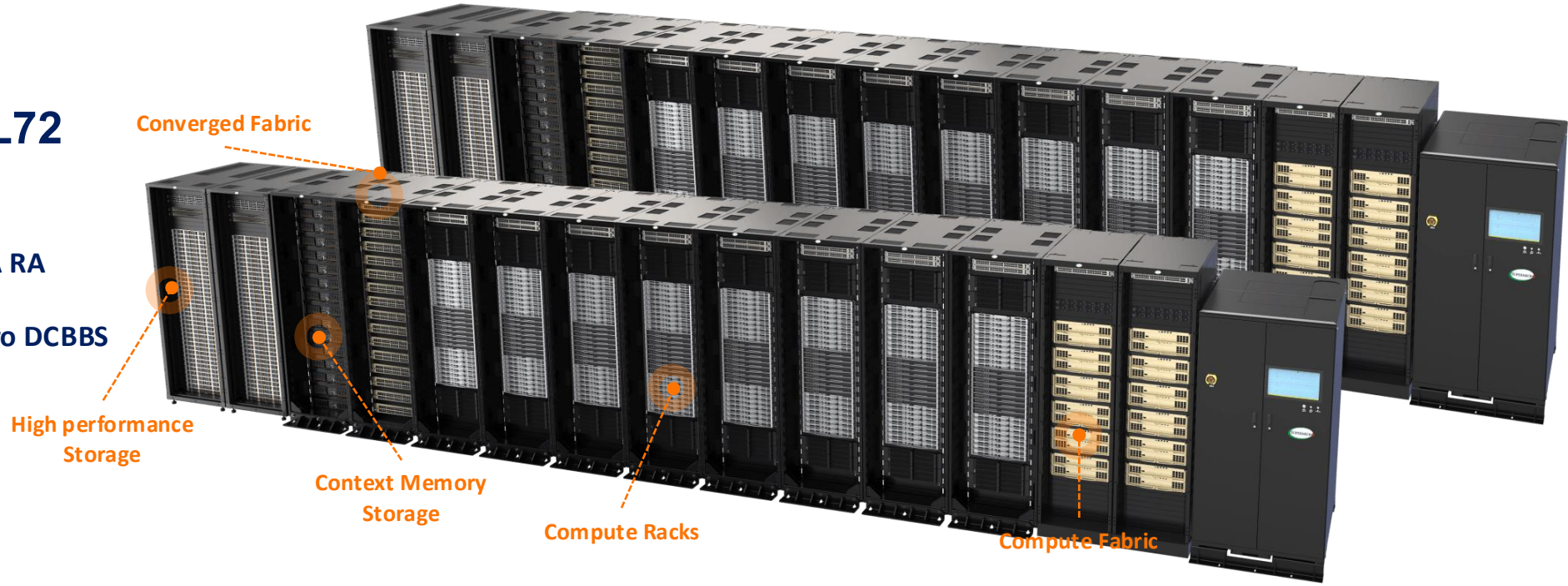
Maintenance Service

MODULAR DESIGN

MODULAR DESIGN - CLUSTER

NVIDIA Vera Rubin NVL72

- Scale form 5MW to 1GW
- Scalable Design based on NVIDIA RA
- 1,152-GPUs per Scalable Unit
- Built & Deployed with Supermicro DCBBS Subsystems
- Management Software Suite





NVIDIA HGX B300 2OU

- 144 GPU Density Per Rack
- 48OU (OCP ORV3)
- 1400A DC Bus Bar
- Estimated Power 245kW (N+2)/Rack
- Built & Deployed with Supermicro DCBBS Subsystems
- Management Software Suite

NVIDIA HGX RUBIN NVL8 SUPERCLUSTER

288 GPUs



1 or 2 CPUs per server | 8 Rubin GPUs per server

(Intel, AMD or NVIDIA Vera CPUs)



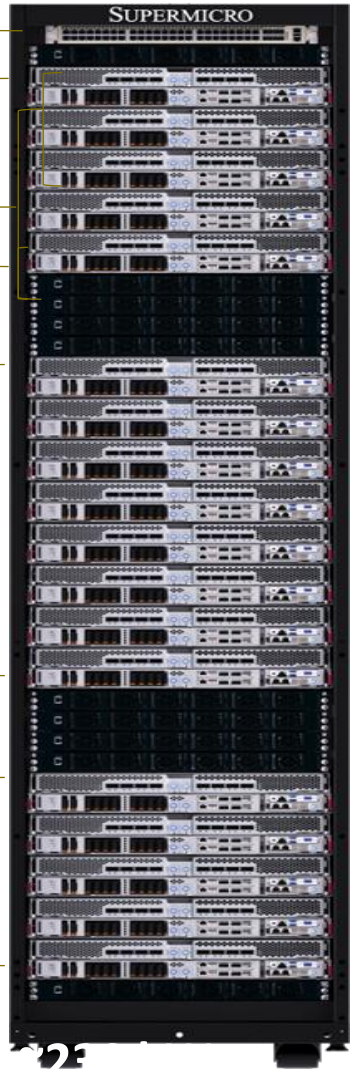
36-node cluster with 288 Rubin GPUs in 4 racks

Compute Nodes & Cooling	E/W Compute Fabric Networking	N/S Scale-Out Networking	Out-of-Band Management
Node: 36 x SYS-222GS-R2BT-ALC **Cooling**: DCBBS direct liquid-cooling stack with multiple CDU options, and L2A sidecar	NVIDIA Quantum-X800 InfiniBand or Spectrum-X Ethernet Switches, up to 800Gb/s	NVIDIA Quantum-X800 InfiniBand or Spectrum-X Ethernet Switches, up to 800Gb/s	NVIDIA Spectrum SN2201 Ethernet switches

Images provided for illustrative purpose, final design will change

ORV RACK SCALE

B300 2OU



1x TOR Switch

1x Power Shelf

5x B300 Servers

4x Power Shelf

8x B300 Servers

4x Power Shelf

5x B300 Servers

1x Power Shelf

~245 kW

FLEX TWIN



2x Power Shelf

44x FlexTwin Servers

2x Power Shelf

~185-220 kW

AMD HELIOS



1x TOR Switch

3x Super Caps Shelf

5x Power Shelf

9x Compute Trays

6x Switch Trays

9x Compute Trays

5x Power Shelf

5x Super Caps Shelf

~235 kW

MGX RACK SCALE

VR NVL72



- **2x** ToR Switches
- **2x** 3U Power Shelves
- **10x** Compute Trays
- **9x** NVLink Switches
- **8x** Compute Trays
- **2x** 3U Power Shelves

~230 kW

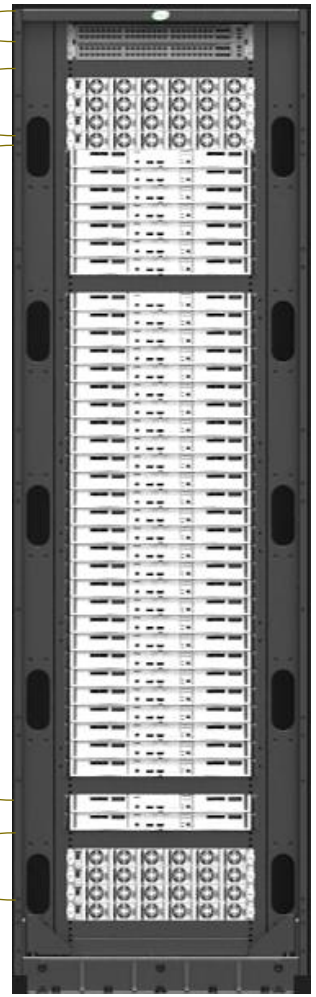
HGX Rubin NVL8



- **1x** ToR Switch
- **2x** 3U Power Shelves
- **9x** 2U Rubin NVL8 Servers
- **2x** 3U Power Shelves

~ 210kW

VR NVL4



- **2x** ToR Switches
- **4x** 1U Power Shelves
- **Up to 36x** VR NVL4 compute trays
- **4x** 1U Power Shelves

~ 362 kW

Vera DLC CPU Rack



- **2x** ToR Switches
- **1x** 3U Power Shelf
- **32x** **100% DLC** Compute Trays
- **2x** 3U Power Shelves

~ 207 kW

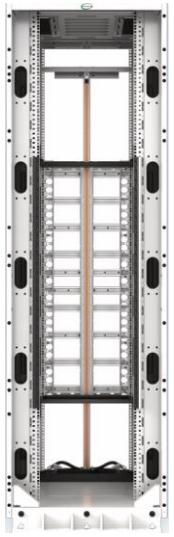
1U 4N Vera, and 1U & 2U Vera C2 Images are provided for illustrative purposes only.

OCP – MGX RACK FRAMES



Height	44 OU	48 OU
Dimensions	29.5 (W) × 47.2 (D) × 88.2 (H) inches	29.5 (W) × 47.2 (D) × 95.7 (H) inches
	750 (W) × 1200 (D) × 2239 (H) mm	750 (W) × 1200 (D) × 2431 (H) mm
Busbars	Included	Included





Height	48 RU	52 RU
Dimensions	29.5/23.6 (W) × 47.2 (D) × 88.2 (H) inches	29.5/23.6(W) × 47.2 (D) × 96.5 (H) inches
	750/600 (W) × 1200 (D) × 2239 (H) mm	750/600 (W) × 1200 (D) × 2450 (H) mm
Busbars	Included	Included

SYSTEM-NETWORKING

SYSTEM



B300

FlexTwin





SYS-421GE-NBRT-LCC
GPU A+ Server



AS -4126GS-NMR-LCC
GPU A+ Server



AS -4126GS-NBR-LCC
GPU A+ Server



AS -4125GS-TNHR2
GPU A+ Server



SYS-422GA-NBRT-LCC
GPU SuperServer



SYS-421GE-NBRT-LCC
GPU SuperServer



SYS-420GP-TNAR+
GPU SuperServer



AS -A126GS-TNBR
GPU A+ Server



SYS-A21GE-NBRT
GPU SuperServer



SYS-A22GA-NBRT
GPU SuperServer



AS -8125GS-TNHR
GPU A+ Server



SYS-822GA-NGR3
AI Training SuperServer



AS -8126S-TNMR
GPU A+ Server



AS -8126S-TNMR
GPU A+ Server



SYS-112C-TN
CloudDC SuperServer



SYS-122C-TN
CloudDC SuperServer



AS -1116CS-TN
CloudDC A+ Server



SYS-222C-TN
CloudDC SuperServer



SSG-122B-NE316R
Storage SuperServer



SSG-222B-NE3X24R
Storage SuperServer

SUPERMICRO

SYSTEM



B300 20U

Key Features

1. 21" OCP ORV3 Rack compatible, 1400A busbar with power shelves, DC-SCM Support
2. Dual Intel® Xeon® 6700 series processors with P-cores, up to 350W TDP
3. NVIDIA HGX B300 8-GPU with 5th Generation NVLink® 1.8TB/s, 2.3TB of HBM3e GPU memory per system
4. 8x NVIDIA ConnectX®-8 SuperNICs, up to 800Gb/s, up to two NICs for North/South traffic
5. 32x DIMMs with up to 8TB at 5200MT/s or up to 4TB at 6400MT/s DDR5 RDIMM
6. 8x Hot-swap E1.S NVMe storage drive bays and 2x M.2 NVMe boot drives
7. 2x 10GbE RJ45 via AIOM card



FlexTwin 1OU

Key Features

Two systems (nodes) in a 1 OU form factor. Each node supports the following:

1. 21" OCP ORV3 Rack compatible, 1400A busbar with power shelves
2. Dual Intel® Xeon® 6900 Series Processors Dual Socket or AMD EPYC™ 9006 SP7 Series Processors
3. Up to 24DIMMs supporting up to 6TB 8000MT/s DDR5 RDIMM or 3TB 8800MT/s DDR5 MRDIMM or up to 32 DDR5 RDIMMs/MRDIMMs (AMD)
4. Flexible networking with up to 1 OCP 3.0 compatible AIOM slots
5. Optional 2 PCIe 5.0 x16 FHHL slot + Optional 1 PCIe 5.0 x16 LP slot
6. Optional 2 E1.S Slots + Optional 2 M.2 NVME drive Slots

NETWORKING



SSE-G3748

48×1G RJ45
6×25/10G SFP28

SONiC

198 Gb/s

1RU



SSE-F3648

48×25/10G SFP28
8×100/40G QSFP28

SONiC

2.0 Tb/s

1RU



SSE-C4632

32×100G QSFP28
breakout to 128×25G

SONiC

3.2 Tb/s

1RU



SSE-T7132

32×400G QSFP-DD

SONiC

12.8 Tb/s

1RU



SSE-T8032

32× dual-port OSFP
(64×400G)

SONiC

25.6 Tb/s

1RU



SSE-T8164

64×800G OSFP

SONiC

51.2 Tb/s

2RU



SSE-T8196

64×400G QSFP112 +
32×800G OSFP

SONiC

51.2 Tb/s

3RU

OOB / BMC
MANAGEMENT

DATA CENTER
10/25/100G leaf and aggregation

400G CLOUD DC / AI / STORAGE
Leaf, spine, and aggregation

800G AI FABRIC / STORAGE
Leaf, spine, and super-spine

1G → 25G → 100G → 400G → 800G

NETWORKING ROADMAP



SSE-G3850

TARGET GA • CQ4/26

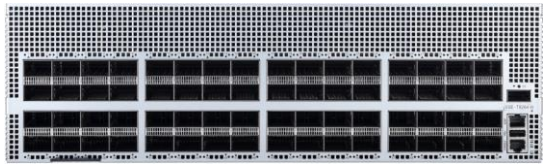
44×1G RJ45
2×100G QSFP28 + 6× sensor RJ45
Rack management

SONiC • RMC OS

244 Gb/s

1RU, EIA/ MGX/ ORV3

OOB / RMM
IPMI, RMC and sensor monitoring



SSE-T8264

TARGET GA • CQ4/26

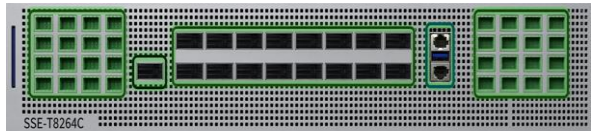
64×1600G OSFP
Pluggable optics

SONiC

102.4 Tb/s

3RU, EIA/ MGX/ ORV3

1600G AI FABRIC
Leaf, Spine



SSE-T8264C

TARGET GA • CQ1/27

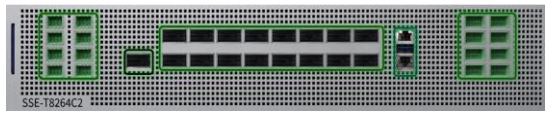
128×800G DR8 MMC-12
Co-packaged optics

SONiC

102.4 Tb/s

2OU, ORV3

CPO AI FABRIC • ORV3
Leaf, Spine



SSE-T8264C2

TARGET GA • CQ1/27

64×1600G DR8 MMC-16
Co-packaged optics

SONiC

102.4 Tb/s

2RU, MGX

CPO AI FABRIC • MGX
Leaf, Spine

DLC COOLING

AI THERMAL CHALLENGE

The cooling architecture must scale with the compute architecture.

Now let's solve the thermal challenge

- Room air is no longer the primary heat-transfer mechanism.
- The question is not “Which CDU?” — it is “What thermal architecture fits this deployment?”
- That architecture spans the server, rack, facility distribution and heat rejection.

AI Rack Thermal Architecture

NVIDIA GB300 NVL72	NVIDIA Vera Rubin NVL72	NVIDIA HGX Rubin NVL8	NVIDIA Vera Rubin NVL4	NVIDIA Vera CPU
~136kW Rack	~230kW Rack	~227kW Rack	~362kW Rack	~207kW Rack



Customer outcome: High-density compute with a predictable, scalable cooling path

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LIQUID COOLING

High-Density Rack	Multi-Rack/POD	No/Limited Facility Water	Mixed Air + Liquid	Campus/Multi-MW
In-Rack CDU Capacity up to 250kW	In-Row CDU Options; 1.8, 2.5, 5MW	Sidecar L2A From 200 to 500kW	RDHx + DLC AC/DC 100,50,20kW	Cooling + Pump Skid
Rack-level distribution and heat transfer	Centralize cooling and scale by pod	Liquid-to-air heat rejection for constrained / retrofit sites	Manage residual room heat and bridge brownfield environments	Facility-scale heat rejection and distribution
				

Match the solution to rack density, facility conditions, site type and scale

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THERMAL DISTRIBUTION LAYER

In-Rack CDU

Minimize piping distance and localize thermal management at the rack

In-Row CDU

Simplify multi-rack distribution and create a repeatable POD building block

Sidecar L2A CDU

Decouple deployment from facility chilled-water availability

Data Center Deployment Decision Tree



Latest Generation DLC GPU Racks

Is Your Data Center Liquid Cooled?







Rack or Row CDU
1,800kW/250kW
Up to 45°C (113°F) warm water

Water Tower or Dry Tower

Is Your Data Center Air Cooled?



L2A Sidecar (up to 500kW) or DC Chiller Doors
Allows DLC racks to operative natively in air-cooled data centers

One portfolio, multiple paths to deployment

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SIMPLIFIED FRAMEWORK

Use the same questions across greenfield, retrofit, and expansion opportunities

LOAD

What is the rack power density?
How many racks / MW?

FACILITY

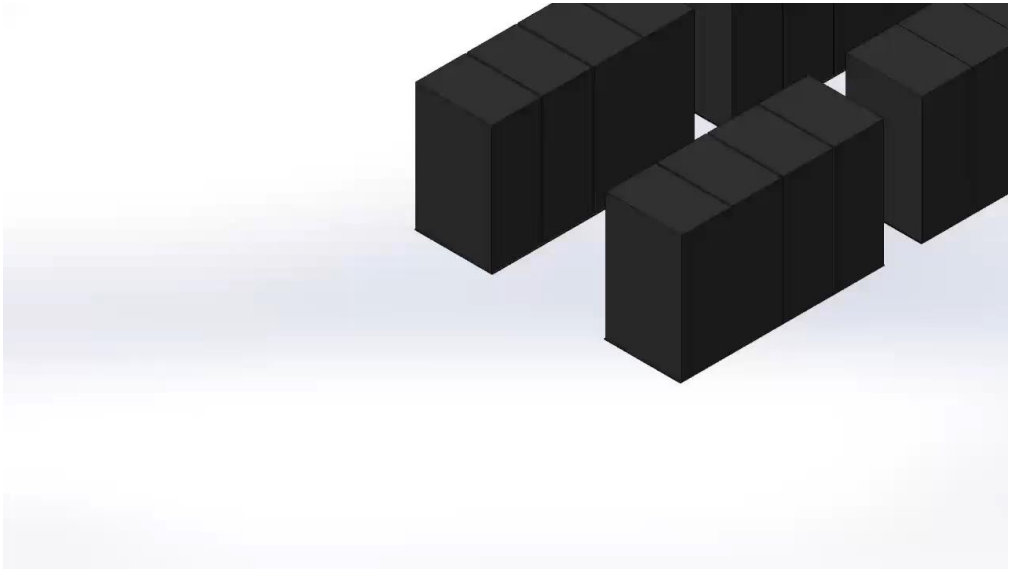
Is facility water available?
At what temperature and flow?

REDUNDANCY

N, N+1 or 2N?
At rack, POD or facility level?

DEPLOYMENT

Greenfield or retrofit?
What is the schedule?



Outcome: Identify the thermal architecture first; select the CDU and supporting infrastructure second.

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RACK – AI FACTORY SCALING

Standardize the thermal building block, then replicate it across pods and data halls

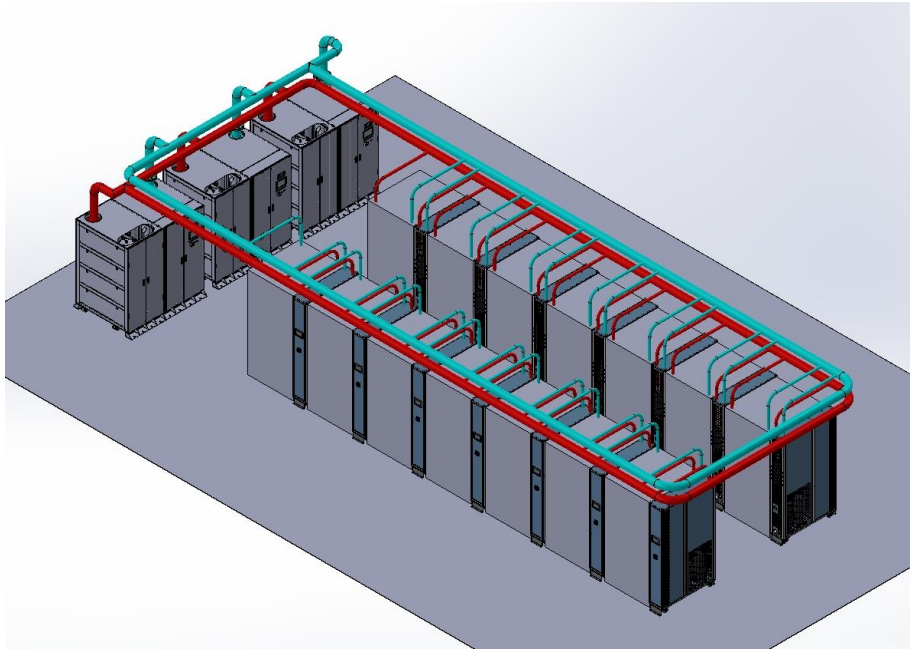
RACK
GPU/CPU
Cold plates
Manifold

POD
CDU × 3
2 Active + 1
Redundant

SFN / TCS
Facility-side
thermal
distribution

PUMP SKID
Flow /
pressure
management

HEAT
REJECTION
Tower / Dry
Cooler



N+1 is not only a CDU count — redundancy must be defined at the correct thermal boundary

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AMD HELIOS OR NVIDIA VERA RUBIN

Latest-generation GPUs double the rack density and the CDU requirement vs prior generation



500kW Sidecar



5MW In-Row CDU



20kW DC Chiller Door

Supported by a New Generation of DCBBS

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DCBBS



Servers
Storage



Network
1G-1.6T



Rack



Power
Shelves



Cooling
In-rack



Cooling
In-Row



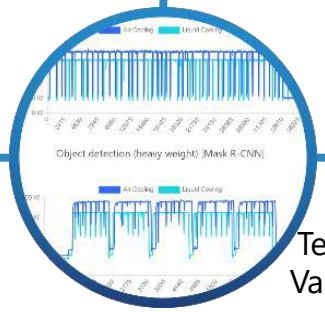
Cooling
RDHx



Cooling
Tower



Solution
Integration



Test and
Validation



On-site
Deployment



Director
Suite

SUPERMICRO

Magelli Roxas

Sr Product Manager-Rack

Supermicro

Magelliroxas@supermicro.com

Steven Huang

Product Manager

Supermicro

Steven.h@supermicro.com

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